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```
> # Load Required Packages
> required_packages <- c("ggplot2","caret","randomForest","e1071","nnet")
> for(pkg in required_packages){
+   if(!require(pkg, character.only = TRUE)){
+     install.packages(pkg)
+     library(pkg, character.only = TRUE)
+   }
+ }
Loading required package: nnet
> set.seed(123)
> Error_Type <- c("AI Bias","Prediction Error","Data Quality Error")
> Frequency <- c(32,38,30)
> data_desc <- data.frame(Error_Type,Frequency)
> data_desc$Percentage <- (data_desc$Frequency /
+                          sum(data_desc$Frequency))*100
> cat("=== DESCRIPTIVE DATA TABLE ===\n")
=== DESCRIPTIVE DATA TABLE ===
> print(data_desc)
          Error_Type Frequency Percentage
1             AI Bias        32         32
2   Prediction Error        38         38
3 Data Quality Error        30         30
> cat("\nTotal Errors:", sum(data_desc$Frequency))

Total Errors: 100
> cat("\nMean Errors:", mean(data_desc$Frequency))

Mean Errors: 33.33333
> cat("\nStandard Deviation:", sd(data_desc$Frequency),"\n")

Standard Deviation: 4.163332
> barplot(data_desc$Frequency,
+         names.arg = data_desc$Error_Type,
+         col = c("steelblue","orange","forestgreen"),
+         main = "AI Error Distribution",
+         ylab = "Frequency")
> pie(data_desc$Frequency,
+     labels = paste(data_desc$Error_Type,
+                    round(data_desc$Percentage,2),"%"),
+     col = c("steelblue","orange","forestgreen"),
+     main = "AI Error Percentage")
> cat("\n=== Chi-Square Test ===\n")

=== Chi-Square Test ===
> print(chisq.test(data_desc$Frequency))

        Chi-squared test for given probabilities
```

```
> ggplot(data_desc,
+        aes(x=Error_Type,y=Frequency,fill=Error_Type)) +
+   geom_bar(stat="identity") +
+   theme_minimal() +
+   ggtitle("AI Error Distribution (ggplot)")
> write.csv(data_desc,
+           "AI_Error_Descriptive.csv",
+           row.names = FALSE)
> n <- 300
> Data_Quality <- rnorm(n, mean=70, sd=10)
> Model_Accuracy <- rnorm(n, mean=75, sd=8)
> Training_Data_Size <- rnorm(n, mean=500, sd=100)
> Processing_Time <- rnorm(n, mean=20, sd=5)
> System_Type <- sample(c("AI Only","Hybrid"),
+                       n, replace=TRUE)
> Error_Type <- ifelse(Data_Quality < 65 &
+                        Processing_Time > 22,
+                      "Prediction Error",
+                      ifelse(Model_Accuracy < 70,
+                             "AI Bias",
+                             "Data Quality Error"))
> data_ml <- data.frame(
+   Data_Quality,
+   Model_Accuracy,
+   Training_Data_Size,
+   Processing_Time,
+   System_Type,
+   Error_Type
+ )
> # Convert to factor
> data_ml$Error_Type <- as.factor(data_ml$Error_Type)
> data_ml$System_Type <- as.factor(data_ml$System_Type)
> trainIndex <- createDataPartition(data_ml$Error_Type,
+                                   p=0.8,
+                                   list=FALSE)
> trainData <- data_ml[trainIndex,]
> testData <- data_ml[-trainIndex,]
> rf_model <- randomForest(Error_Type ~ .,
+                          data=trainData,
+                          ntree=200)
> rf_pred <- predict(rf_model, testData)
> cat("\n=== Random Forest Results ===\n")

=== Random Forest Results ===
> print(confusionMatrix(rf_pred, testData$Error_Type))
Confusion Matrix and Statistics
```

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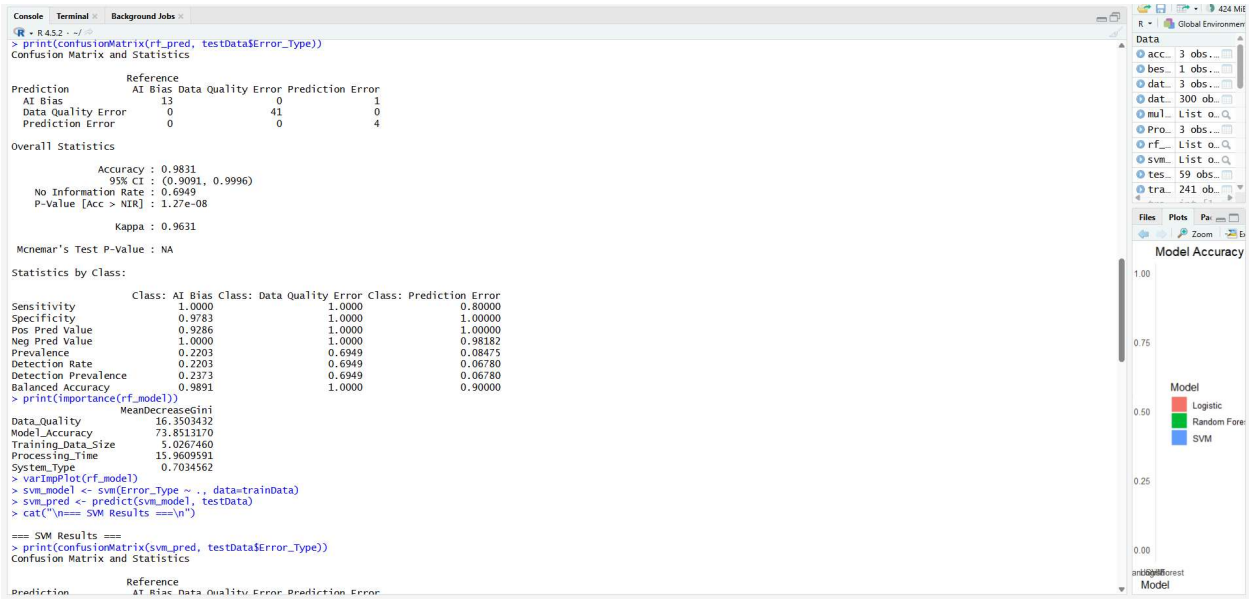

```
> print(confusionMatrix(rf_pred, testData$Error_Type))
Confusion Matrix and Statistics

                    Reference
Prediction           AI Bias Data Quality Error Prediction Error
  AI Bias             13                   0                 1
  Data Quality Error   0                  41                 0
  Prediction Error     0                   0                 4

Overall Statistics

               Accuracy : 0.9831
                 95% CI : (0.9091, 0.9996)
    No Information Rate : 0.6949
    P-Value [Acc > NIR] : 1.27e-08

                  Kappa : 0.9631

 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: AI Bias Class: Data Quality Error Class: Prediction Error
Sensitivity                 1.0000                    1.0000                 0.80000
Specificity                 0.9783                    1.0000                 1.00000
Pos Pred Value              0.9286                    1.0000                 1.00000
Neg Pred Value              1.0000                    1.0000                 0.98182
Prevalence                  0.2203                    0.6949                 0.08475
Detection Rate              0.2203                    0.6949                 0.06780
Detection Prevalence        0.2373                    0.6949                 0.06780
Balanced Accuracy           0.9891                    1.0000                 0.90000
> print(importance(rf_model))
                  MeanDecreaseGini
Data_Quality             16.3503432
Model_Accuracy           73.8513170
Training_Data_Size        5.0267460
Processing_Time          15.9609591
System_Type               0.7034562
> varImpPlot(rf_model)
> svm_model <- svm(Error_Type ~ ., data=trainData)
> svm_pred <- predict(svm_model, testData)
> cat("\n=== SVM Results ===\n")

=== SVM Results ===
> print(confusionMatrix(svm_pred, testData$Error_Type))
Confusion Matrix and Statistics

                    Reference
Prediction           AI Bias Data Quality Error Prediction Error
```


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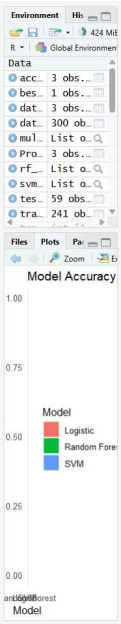
```
Source
Console  Terminal  Background Jobs
R 4.5.2 · ~/
 Prediction Error          0                    0                    2

Overall Statistics

               Accuracy : 0.9153
                 95% CI : (0.8132, 0.9719)
    No Information Rate : 0.6949
    P-Value [Acc > NIR] : 4.799e-05

                  Kappa : 0.8102

 Mcnemar's Test P-Value : 0.3916

Statistics by Class:

                     Class: AI Bias Class: Data Quality Error Class: Prediction Error
Sensitivity                 0.9231                    0.9756                  0.40000
Specificity                 0.9348                    0.8889                  1.00000
Pos Pred Value              0.8000                    0.9524                  1.00000
Neg Pred Value              0.9773                    0.9412                  0.94737
Prevalence                  0.2203                    0.6949                  0.08475
Detection Rate              0.2034                    0.6780                  0.03390
Detection Prevalence        0.2542                    0.7119                  0.03390
Balanced Accuracy           0.9289                    0.9322                  0.70000
> multi_model <- multinom(Error_Type ~ .,
+                        data=trainData,
+                        trace=FALSE)
> multi_pred <- predict(multi_model, testData)
> cat("\n=== Logistic Regression Results ===\n")

=== Logistic Regression Results ===
> print(confusionMatrix(multi_pred, testData$Error_Type))
Confusion Matrix and Statistics

                      Reference
Prediction             AI Bias Data Quality Error Prediction Error
  AI Bias                   13                  0                 2
  Data Quality Error         0                 39                 1
  Prediction Error           0                  2                 2

Overall Statistics

               Accuracy : 0.9153
                 95% CI : (0.8132, 0.9719)
    No Information Rate : 0.6949
    P-Value [Acc > NIR] : 4.799e-05

                  Kappa : 0.8186
```




```
Source
Console  Terminal  Background Jobs
R 4.5.2 · ~/
 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: AI Bias Class: Data Quality Error Class: Prediction Error
Sensitivity                 1.0000                    0.9512                  0.40000
Specificity                 0.9565                    0.9444                  0.96296
Pos Pred Value              0.8667                    0.9750                  0.50000
Neg Pred Value              1.0000                    0.8947                  0.94545
Prevalence                  0.2203                    0.6949                  0.08475
Detection Rate              0.2203                    0.6610                  0.03390
Detection Prevalence        0.2542                    0.6780                  0.06780
Balanced Accuracy           0.9783                    0.9478                  0.68148
> rf_accuracy <- sum(rf_pred == testData$Error_Type) /
+   nrow(testData)
> svm_accuracy <- sum(svm_pred == testData$Error_Type) /
+   nrow(testData)
> log_accuracy <- sum(multi_pred == testData$Error_Type) /
+   nrow(testData)
> cat("\nRandom Forest Accuracy:", rf_accuracy)

Random Forest Accuracy: 0.9830508
> cat("\nSVM Accuracy:", svm_accuracy)

SVM Accuracy: 0.9152542
> cat("\nLogistic Regression Accuracy:", log_accuracy)

Logistic Regression Accuracy: 0.9152542
> accuracy_data <- data.frame(
+   Model = c("Random Forest","SVM","Logistic"),
+   Accuracy = c(rf_accuracy,svm_accuracy,log_accuracy)
+ )
> ggplot(accuracy_data,
+        aes(x=Model,y=Accuracy,fill=Model)) +
+   geom_bar(stat="identity") +
+   theme_minimal() +
+   ggtitle("Model Accuracy Comparison")
> best_model <- accuracy_data[which.max(accuracy_data$Accuracy),]
> cat("\nBest Model:\n")

Best Model:
> print(best_model)
          Model  Accuracy
1 Random Forest 0.9830508
> write.csv(data_ml,
+           "AI_Error_ML_Dataset.csv",
+           row.names = FALSE)
>
```

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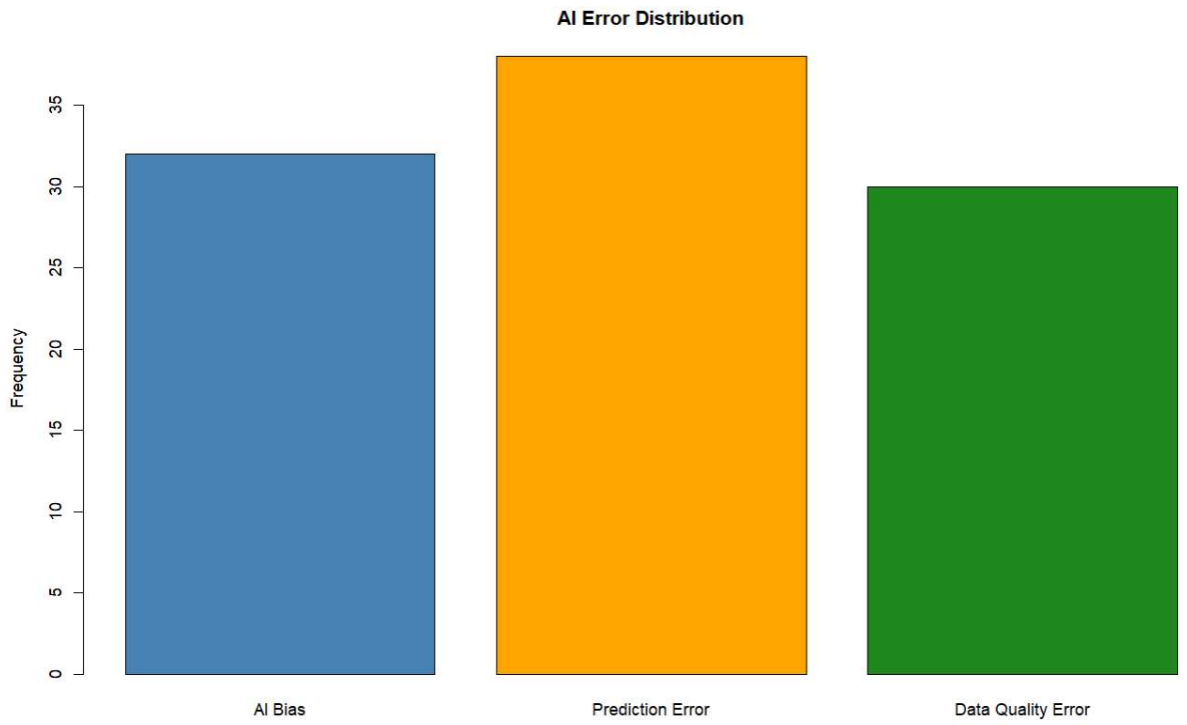

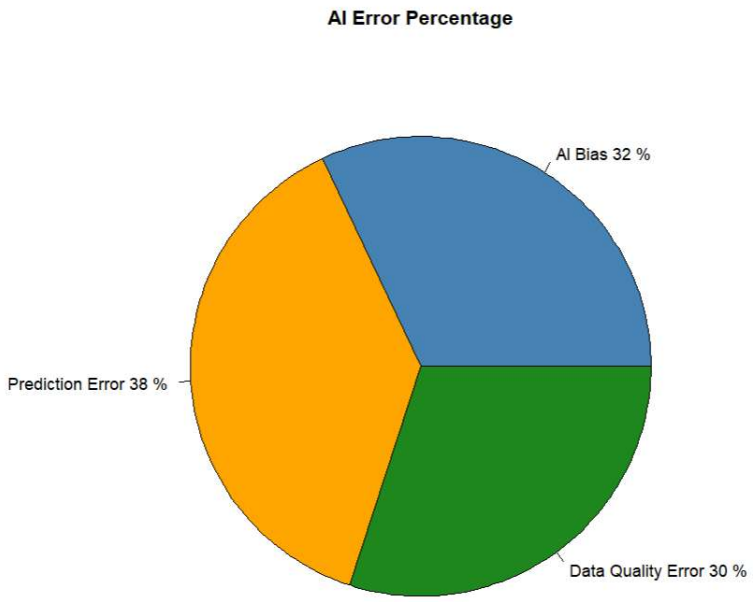

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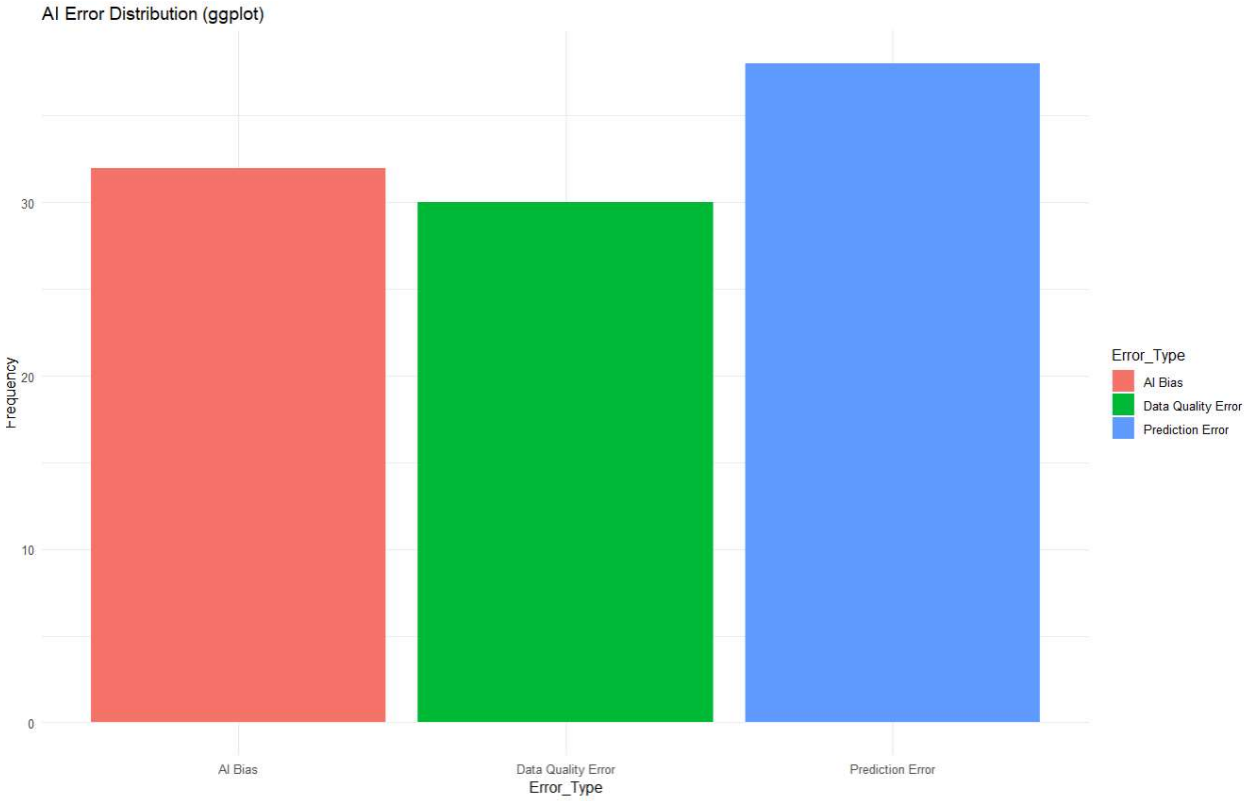

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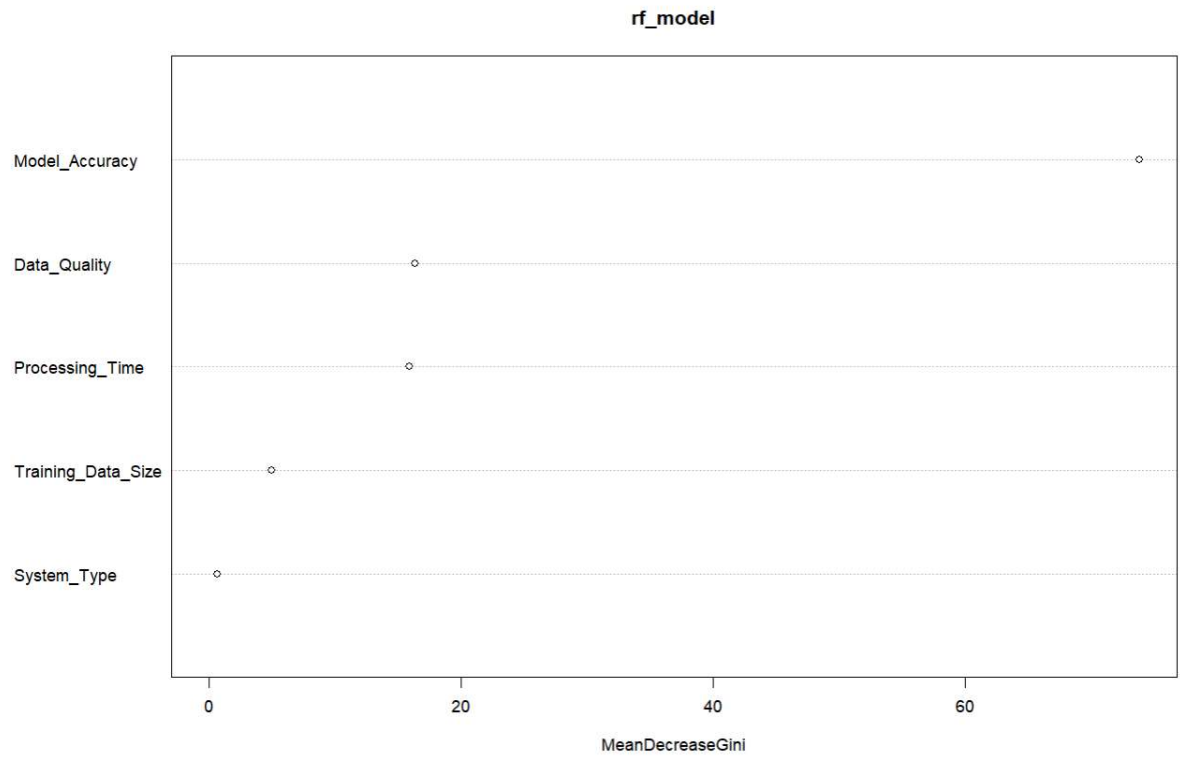
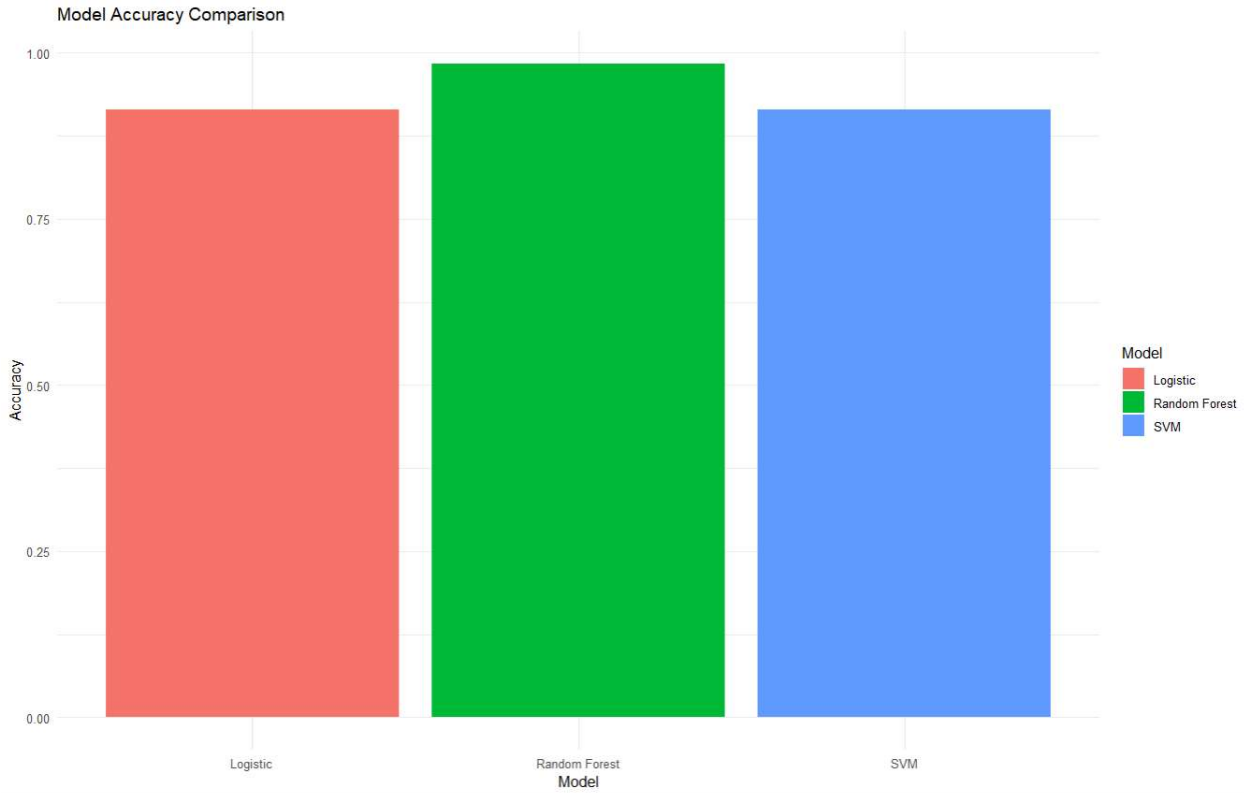

rf_model
Model_Accuracy
Data_Quality
Processing_Time
Training_Data_Size
System_Type
0
20
40
60
MeanDecreaseGini


Model Accuracy Comparison
1.00
0.75
0.50
0.25
0.00
Accuracy
Logistic
Random Forest
SVM
Model
Model
Logistic
Random Forest
SVM